

CipherStream-128X: Hardware Accelerator Documentation

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1. Design Document

1.1. Pipeline Architecture

To eliminate internal processing bottlenecks, CipherStream-128X utilizes a fully unrolled 10-stage pipeline. Unlike iterative designs that reuse a single hardware block over multiple clock cycles, this architecture physically instantiates each of the 10 AES rounds (SubBytes, ShiftRows, MixColumns, AddRoundKey) as a distinct hardware stage. This spatial parallelism allows ten separate 128-bit data blocks to reside in the pipeline simultaneously at varying stages of encryption. Once the pipeline is primed, the internal AES engine is capable of retiring one full 128-bit block per clock cycle.

1.2. S-Box Implementation Choice

The SubBytes transformation is the most computationally heavy operation in the AES algorithm. In this design, the S-Boxes were implemented using pre-computed Look-Up Tables (LUTs) rather than calculating the Galois Field multiplicative inverse on-the-fly. This approach leverages the fast, distributed RAM/LUT structure of the FPGA fabric. While it consumes a larger physical area, it drastically minimizes combinational path delay, allowing the core to achieve a significantly higher maximum frequency (F_{max}).

1.3. Key Schedule Strategy

The accelerator employs an **On-the-Fly Key Expansion** strategy heavily coupled with the unrolled datapath. The round keys are generated in parallel as the data progresses through the pipeline. To support dynamic key updates without corrupting in-flight data, the key state is registered and shifted alongside its corresponding data block. This guarantees that new keys injected via the AXI bus only apply to plaintext entered *after* the key update.

1.4. Interface and Buffering Design

The IP core is integrated into the SoC environment using a dual AXI4-Lite interface strategy:

- **Control (AXI4-Lite Slave):** Manages the 128-bit Key setup (mapped across four 32-bit registers) and exposes the Start, Stop, and Status (Busy, Idle, Done) flags. The Done signal is correctly implemented as a 1-cycle pulse upon completion of an encryption burst.
- **Data (AXI4-Lite Master):** Handles the automated fetching and write-back of data to the memory subsystem. Because the pipeline evaluates continuously, a "Catching Register" acts as an output buffer, holding the valid ciphertext stable while the Master FSM executes the multi-cycle write back to memory.

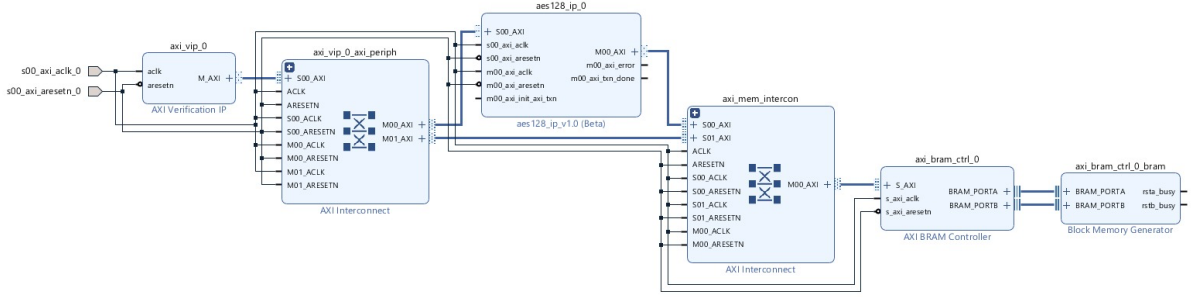


Figure 1: Vivado Block Design demonstrating the integration of the CipherStream-128X IP via AXI4-Lite Master and Slave interfaces.

1.5. Tradeoff Analysis

- **Area vs. Throughput:** An unrolled 10-stage pipeline consumes roughly 10x the Slice logic of an iterative design. However, this tradeoff was intentionally made to prioritize maximum gigabit throughput for continuous data streams.
- **Bandwidth vs. Pipeline Stalls:** While the internal pipeline is stall-free, the system-level integration utilizes a 32-bit AXI4-Lite interface. Due to the bandwidth mismatch between the 128-bit datapath and the 32-bit bus, intentional "Communication Stalls" are implemented. The AXI Master executes four discrete 32-bit read transactions to assemble a single plaintext block before asserting `pt_valid`, ensuring protocol compliance and data integrity across the memory-mapped boundary.

2. Functional Verification Results

To guarantee strict adherence to the FIPS-197 standard and AXI protocol rules, the IP core was verified using a comprehensive SystemVerilog testbench utilizing the Xilinx AXI Verification IP (VIP).

- **Standardized AES Test Vectors:** The accelerator successfully passed all re-

quired NIST test vectors, specifically FIPS-197 Appendix B, Appendix C, and the Zero-Key/Zero-Text edge case.

- **Randomized Verification:** A 25-iteration randomized stress test confirmed 100% retirement without pipeline stalls, deadlocks, or in-flight key corruption.
- **AXI Protocol Validation:** The design successfully verified 16-cycle synchronous reset handling and valid/ready backpressure handshake compliance across the Master and Slave boundaries.

3. Synthesis Report

The design was synthesized out-of-context (OOC) targeting the Xilinx Artix-7 (xc7a200t-fbv676-2) FPGA to accurately measure the internal logic performance.

- **Logic Utilization (Slice LUTs):** 10,490
- **Register Utilization (Slice FFs):** 3,082
- **Block RAM (BRAM) Usage:** 0 (S-Boxes implemented in distributed LUT logic)
- **Target Clock Period:** 10 ns (100 MHz)
- **Worst Negative Slack (WNS):** 1.574 ns
- **Maximum Frequency (F_{max}):** 118.68 MHz

(Note: F_{max} is calculated as $1000/(10.000 - 1.574)$)

4. Performance Analysis

4.1. Latency

Due to the fully unrolled architecture, the processing latency of the AES core itself is strictly deterministic. It takes exactly **10 clock cycles** for a single 128-bit plaintext block to traverse from the input register to the valid ciphertext output register.

4.2. Throughput

The theoretical peak throughput of the internal AES hardware engine is calculated based on its ability to process 128 bits per clock cycle at the maximum operating frequency:

$$\text{Throughput} = F_{max} \times 128 \text{ bits/cycle}$$

$$\text{Throughput} = 118.68 \text{ MHz} \times 128 \text{ bits} = 15,191.04 \text{ Mbps}$$

The peak internal hardware throughput is **15.19 Gbps**. (Note: Effective system-level throughput will be scaled by the 32-bit AXI4-Lite bus fetch cycles).